Laplace's equation

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March 3, 2026

1 Statement of the problem

Solve Laplace's equation

$$\nabla^2 V = 0 \tag{1}$$

where V is the electrostatic potential.

The electrostatic problem is defined in Figure (1) where we define a solution region.

Results from using FE is shown in Figure (2).

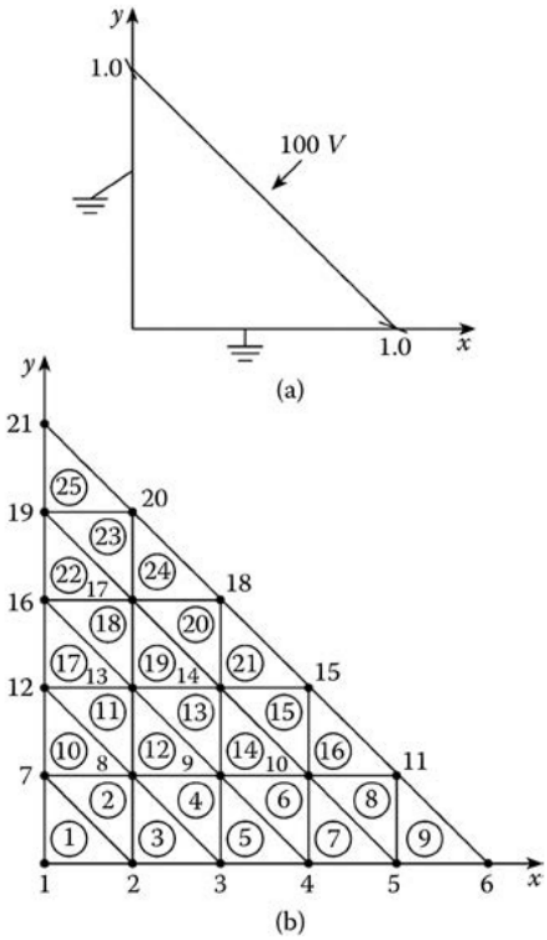

Figure 1: (a) is electrostatic problem and (b) is the solution region

Node	X	Y	Potential
1	0.00	0.00	0.000
2	0.20	0.00	0.000
3	0.40	0.00	0.000
4	0.60	0.00	0.000
5	0.80	0.00	0.000
6	1.00	0.00	50.000
7	0.00	0.20	0.000
8	0.20	0.20	18.182
9	0.40	0.20	36.364
10	0.60	0.20	59.091
11	0.80	0.20	100.000
12	0.00	0.40	0.000
13	0.20	0.40	36.364
14	0.40	0.40	68.182
15	0.60	0.40	100.000
16	0.00	0.60	0.000
17	0.20	0.60	59.091
18	0.40	0.60	100.000
19	0.00	0.80	0.000
20	0.20	0.80	100.000
21	0.00	1.00	50.00

Number of nodes = 21, number of elements = 25, number of fixed nodes = 15.

Figure 2: Potential computed from using FE